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Experiment No 8

Aim: Write a shell script to perform arithmetic operations.

LO5: Implement basic shell scripts for different applications.

Theory:

Shell scripting is a powerful tool in Unix/Linux environments that allows automation of tasks by executing commands sequentially. Arithmetic operations in shell scripts are performed using built-in operators and external tools like expr or bc.

Arithmetic Operators in Shell Scripting:

1. Addition (+): Adds two numbers.
2. Subtraction (-): Subtracts one number from another.
3. Multiplication (*): Multiplies two numbers.
4. Division (/): Divides one number by another.
5. Modulus (%): Finds the remainder when one number is divided by another.
6. Exponentiation ()**: Raises a number to the power of another.

Methods to Perform Arithmetic in Shell Scripts:

- Using expr command: This is a traditional method used in older shell scripts.
  Example: sum=`expr $a + $b`
- Using let command: This allows performing arithmetic calculations.
  Example: let sum=a+b
- Using double parentheses (( )): This is a modern approach in Bash.
  Example: sum=$((a + b))
- Using bc command: Used for floating-point arithmetic.
  Example: echo "scale=2; $a / $b" | bc

Shell Script:

Open | exp8.sh ~/Desktop | Save

```
#!/bin/bash
echo "Enter first number:"
read num1
echo "Enter second number:"
read num2
echo "Choose operation : + - * /"
read op

case $op in
     +)
        result=$((num1+num2))
        echo "Result : $result"
        ;;
     -)
        result=$((num1-num2))
        echo "Result : $result"
        ;;
     "*")
        result=$((num1*num2))
        echo "Result : $result"
        ;;
     /)
        result=$((num1/num2))
        echo "Result : $result"
        ;;
     *)
        echo "Invalid operation"
        ;;
 esac
```

Output:

```
student@student-VirtualBox:~/Desktop$ gedit exp8.sh
student@student-VirtualBox:~/Desktop$ chmod +x exp8.sh
student@student-VirtualBox:~/Desktop$ ./exp8.sh
Enter first number:
12
Enter second number:
23
Choose operation : + - * /
+
Result : 35
student@student-VirtualBox:~/Desktop$ chmod +x exp8.sh
student@student-VirtualBox:~/Desktop$ ./exp8.sh
Enter first number:
12
Enter second number:
23
Choose operation : + - * /
-
Result : -11
student@student-VirtualBox:~/Desktop$ chmod +x exp8.sh
student@student-VirtualBox:~/Desktop$ ./exp8.sh
Enter first number:
12
Enter second number:
23
Choose operation : + - * /
*
Result : 276
```

Conclusion:

In this experiment, we successfully created a shell script to perform basic arithmetic operations. We learned different methods to implement arithmetic in shell scripting, including expr, let, (( )), and bc. This script is useful for automating calculations in Unix/Linux systems.

LO5 is achieved.

PO's achieved – PO1, PO5, PO8, PO9, P010, PO12